

EXPERIMENT 1

Title: Construct and Verify the Performance of Bi-Level and Multi-Level Confusion Matrix using Python

✓ AIM

To construct **bi-level (binary)** and **multi-level (multi-class)** confusion matrices using Python and to **verify their performance** by calculating accuracy.

✓ ALGORITHM

Algorithm for Constructing a Confusion Matrix (Manual Method)

1. **Start**
2. Define the actual class labels (y_{true}).
3. Define the predicted class labels (y_{pred}).
4. Create a list containing all unique labels (e.g., [0,1] for bi-level and [0,1,2] for multi-level).
5. Initialize a matrix of size $N \times N$, where N is the number of labels.
6. For each pair of actual and predicted values:
 - Find the row index corresponding to actual label.
 - Find the column index corresponding to predicted label.
 - Increment the matrix cell at that index.
7. Display the confusion matrix.
8. Count the number of correctly predicted values.
9. Calculate **Accuracy = (Correct Predictions / Total Samples)**.
10. Display the accuracy value.
11. **Stop**

✓ INPUT

Bi-Level Input

```
12. Actual (y_true): [0, 1, 1, 0, 1, 0]
13. Predicted (y_pred): [0, 1, 0, 0, 1, 1]
14. Labels: [0, 1]
```

Multi-Level Input

```
15. Actual (y_true): [0, 1, 2, 1, 0, 2]
16. Predicted (y_pred): [0, 2, 2, 1, 0, 1]
17. Labels: [0, 1, 2]
```

✓ OUTPUT

```
main.py  [ ] [ ] [ ] Share Run Output Clear

1 # Function to create confusion matrix manually
2 def confusion_matrix_manual(y_true, y_pred, labels):
3     size = len(labels)
4     cm = [[0]*size for _ in range(size)]
5
6     label_to_index = {label: i for i, label in enumerate
                        (labels)}
7
8     for t, p in zip(y_true, y_pred):
9         cm[label_to_index[t]][label_to_index[p]] += 1
10    return cm
11
12 # Function to calculate accuracy manually
13 def accuracy_manual(y_true, y_pred):
14     correct = sum(t == p for t, p in zip(y_true, y_pred))
15     return correct / len(y_true)
16
17 # -----
18 # 1. BI-LEVEL (Binary) Example
19 # -----
20 y_true_binary = [0, 1, 1, 0, 1, 0]
21 y_pred_binary = [0, 1, 0, 0, 1, 1]
22
23 labels_binary = [0, 1]
```

```
=== Bi-Level Confusion Matrix (Manual) ===
[2, 1]
[1, 2]
Accuracy: 0.6666666666666666

=== Multi-Level Confusion Matrix (Manual) ===
[2, 0, 0]
[0, 1, 1]
[0, 1, 1]
Accuracy: 0.6666666666666666

=== Code Execution Successful ===
```

✓ RESULT

The bi-level and multi-level confusion matrices were successfully constructed using Python.

The performance of both models was verified by manually calculating the **accuracy**.